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Methods Article

CATHe2: Enhanced CATH superfamily detection using ProstT5 and structural alphabets

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Abstract

The CATH database is a free publicly available online resource that provides annotations about the evolutionary and structural relationships of protein domains. Due to the flux of protein structures coming mainly from the recent breakthrough of AlphaFold and therefore the non-feasibility of manual intervention, the CATH team recently developed an automatic CATH superfamily (SF) classifier called CATHe, which uses a feed-forward neural network (FNN) classifier with protein Language Model (pLM) embeddings as input. Using the same dataset of remote homologues (with a 20% sequence identity threshold), this paper presents CATHe2, which improves on CATHe by switching the old pLM ProtT5 for one of the most recent versions called ProstT5, and by incorporating domain 3D information into the classifier through Structural Alphabet representation, specifically, 3Di sequence embeddings. Finally, CATHe2 implements a new version of the FNN classifier architecture, fine-tuned to perform at the CATH superfamily prediction task. The best CATHe2 model reaches an accuracy of 92.2% ± 0.7% with an F1 score of 82.3% ± 1.3%, which constitutes an improvement of 9.9% on the F1 score and 6.6% on the accuracy, from the previous CATHe version (85.6% ± 0.4% accuracy and 72.4% ± 0.7% F1 score) on its largest dataset (∼1700 superfamilies). This model uses ProstT5 amino acid (AA) sequence and 3Di sequence embeddings as input to the classifier, but a simplified version requiring only AA sequences, already improves CATHe's F1 score by 6.7% ± 1.3% and accuracy by 6.6% ± 0.7% on its largest dataset.

Keywords: CATH; CATHe; Protein Language Model (pLM); ProstT5; Structural Alphabet; 3Di; protein; classifier

Introduction

Proteins are essential to life, forming the foundation of virtually all biological processes. They are complex nanomachines with specific functions which largely depend on their unique 3D structure. Figuring out what shapes proteins fold into is known as the "protein folding problem" and has stood as a grand challenge in biology since 1972 [1, 2]. Alphafold2 is a 2020 automated algorithm which reached experimental performance in the protein folding problem, which is now considered solved for a large part of known proteins [3]. Since then, a database of AlphaFold inferred 3D protein conformations also called CSM (Computed Structure Model) has been filled with over 200 million protein structure predictions. This database is the AlphaFold database or AFDB for short [4]. As protein evolution gives rise to families of structurally related proteins, within which sequence identities can be extremely low, structure-based classifications are needed to identify unanticipated relationships in known structures, now present in overwhelming numbers. CATH is a classification system that fills this need and even allows inference of domain function when paired with FunFam subclassification [5]. The corresponding CATH database is updated daily with the release of a snapshot version of the most up-to-date classifications. A full-release version called CATH-Plus including a greater depth of information about CATH domains is also uploaded annually, benefiting from ever improving automatic classification algorithms [6–8]. The latest CATH-Plus version available at the time of writing is 4.4.0, and it includes 6,573 superfamilies.

After the major bioinformatics revolution with AlphaFold2 (AF2) [3], bioinformaticians are now witnessing a second revolution with the rise of protein Language Models (pLMs) [9–12]. After the surprising huge leap of performance in NLP (Natural Language Processing) allowed by LLMs (Large Language Models) in recent years, protein processing soon took advantage of this new technique to process Amino Acid (AA) sequences, resulting in automatic and powerful feature extractors such as the ProtT5 pLM and CATHe classifier [13] (in this paper, "ProtT5" refers to ProtT5-XL-U50 from the T5 models presented in the ProtTrans paper [14]). With new pLMs, pLM usage, and larger protein 3D conformation databases, the remote homologue detection and homology based inference (HBI) tool landscape has been rapidly evolving, making use of fast and sensitive large scale remote homologue detection using structural information [15–17] or better feature extraction from primary sequences.

While standard sequence profile and Hidden Markov Model (HMM)-based alignment tools such as MMSeqs2 [18] and HH-suite [19, 20] have been successful, they reach their limits in the twilight or midnight zones of sequence similarity. In contrast,

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structural alignment tools like Foldseek [17] leverage 3D protein information to perform better at detecting remote homology and are significantly faster than earlier 3D alignment methods such as DALI [21]. These techniques are the state of the art in their own category, but the new wave of techniques inspired by pLMs is already being regarded as more promising as they can be faster and more sensitive in some cases. Embedding-based annotation transfer (EAT) transfers protein annotation to a close protein using distance test, like Euclidean distance or cosine similarity between pLM embeddings [22–25]. EAT can be coupled with Contrastive Learning (CL) to extend the distance between embedding distribution clusters and thus improving EAT results [24, 26]. Protein Embeddings Based Alignment (EBA or PEbA) uses pLM embeddings alignment to find homologues and has been very successful, with a lot of work using it to beat former state-of-the-art methods in various domains [27–29]. Protein artificial neural network (ANN) classifiers, such as those used in CATHe and this study, are also valuable tools for homology-based inference (HBI). These classifiers perform HBI by using supervised learning to develop internal representations for homology detection.

All of the previous techniques are based on three possible types of input. First, primary sequences that can be vectorized in many different ways, pLM embeddings is one of them. Second, evolutionary information, including Multiple Sequence Alignment (MSA), (HMM-based techniques use MSA but some pLM-based techniques too [30–33]), and the last input type is structural information, which can take multiple forms like full and precise 3D shape or just contact information, secondary structure annotation, or 1D representation similar to Structural Alphabets (SA) [34]. All this structural information can be inferred information or ground truth and used as is or embedded using pLM for more complex feature extraction [35–38, 25]. Recent structural alignment tools are promising, potentially more meaningful and accurate for HBI since structure is more important for protein function than primary sequences, plus, most pairs of proteins with similar structures populate the midnight zone [39]. Nowadays, many new techniques revolve around finding shortcuts to use 3D information without having to compute the 3D shape of proteins or skipping MSA construction, and thus speeding up potential subsequent structural alignment [40,41]. Using both structural and primary sequence information, i.e., hybrid-information-based methods, has been shown to outperform other categories of methods [42] for remote homology detection and HBI. As primary sequences have been already studied extensively, current research is focused on making the most out of protein 3D structure. Most sensitive and fast remote homology detection approaches today use embedding-based and structure-aware tools such as Foldseek, ESMFold [43] or AlphaFold-Multimer [44], for instance, to produce relevant data for subsequent HBI techniques like EBA, which seems to be one of the most effective. Some techniques like Foldseek-TM [17] use shortcuts to be even faster but less sensitive [27]. However, there is a way to be even faster, without losing sensitivity, i.e., using discriminative methods [9]. Discriminative methods realize HBI tasks without the need for any kind of alignment, as with the ANN classifiers of CATHe and CATHe2. This is why automated protein annotation with ANN classifiers is a good long-term solution for large-scale-specific HBI tasks. Although it is less transparent, ANN classifiers are potentially faster and more accurate as no alignment is required and the classifier learns on its own how to find homologs, whatever the complexity of the homology relation. Given the right amount of representative data, sufficiently complex ANN classifiers can learn the proper way to classify (and annotate) proteins based on their most relevant properties extracted by a trained pLM, benefiting from transfer learning.

In this paper, we introduce CATHe2, a new state-of-the-art classifier (made with the TensorFlow/Keras framework), allowing for large-scale automatic protein domain classification among the CATH superfamilies by testing various protein language models. For CATHe2, both AA sequence embeddings and 3D structure information are used to perform the HBI of CATH annotations with an ANN classifier. The 3D information is also embedded, thanks to the 3Di SA developed for Foldseek and the ProstT5 pLM. The first version of CATHe relied on both pLM AA sequence embedding and ANN classifier usage to improve on the automatic CATH annotation task up to an F1 score of 72.4% on the corresponding dataset. CATHe2 improves on this method, using newer pLMs, adding protein 3D information as input (in the form of SA embeddings), and fine tuning the previous ANN classifier architecture for a final model F1 score of 82.3% on almost the same dataset. But CATHe and CATHe2 are not the only CATH annotation algorithms tested in recent years, many different pLMs were tested with different metrics, across many papers using different techniques and approaches for CATH annotation inference. As shown in Table 1, the current ranking of HBI techniques for CATH annotations evidenced by literature review seems to be classifiers >EBA >EAT+CL >EAT. Similarly, a ranking of pLM tested for this task should resemble ProstT5 >ProtT5 >TM-Vec >Ankh large >ESM2 >ESM1b although this ranking depends strongly on HBI technique. Almost all of these pLMs were also tested for CATHe2.

Table 1. CATH annotation HBI technique performance ranking.

Model + HBI technique	Accuracy
CATHe2 AA + 3Di (ProstT5 full + classifier with AA + 3Di)	92.0%
CATHe2 AA only (ProstT5 full + classifier AA only)	90.5%
CATHe (ProtT5 + classifier)	85.6% [14]
ProstT5 + EBA	85.3% [28]
ProstT5 + EAT + CL	81.0% [27]
ProtT5 + EBA	80.8% [28]
TM-Vec + EAT	80.1% [47] (supplements)
ESM1b + EBA	78.5% [28]
ProstT5 + EAT + CL (3Di)	78.0% [27] (supplements)
ProstT5 + EAT (3Di)	77.0% [27]
ProtT5 + EAT + CL	76.0%–77.75% [24, 45] (supplement)
Ankh large + EAT + CL	75.0% [27] (supplement)
Ankh base + EAT	74.8% [23]
ProstT5 + EAT	73.0% [27]
Ankh large + EAT	71.7% [23]
ProtT5 + EAT	68.0%–71.0% [25, 23]
ESM2 15B + EAT	65.4% [23]
ESM1 + EAT	62.0%–64.5% [27, 23]

This table presents a comprehensive performance review of pLM and HBI technique combinations for CATH annotation inference, ranked from best to worst based on the literature available on this topic [46]. When no input type is mentioned, it means only AA embeddings were used. Classifiers tend to perform better at CATH HBI, with ProstT5 appearing particularly effective for CATH annotation transfer. Although the datasets used vary, to the best of our knowledge, the sequence identity threshold (20%) adopted in CATHe and CATHe2 is the most stringent. Additionally, it is worth noting that, apart from the first two rows (CATHe and CATHe2), the accuracies shown are averages over multiple predicted CATH annotation levels (Class, Architecture, Topology, and Homologous Superfamilies), not just the Superfamily level. This inflates the reported accuracy compared to CATHe and CATHe2, as predicting higher levels—especially Class and Architecture is significantly easier.

Materials and methods

Data and datasets

To conduct as accurate and fair comparison as possible, CATHe2 datasets are based on the CATHe ones [13]. In CATHe, two SF datasets were used. The large SF dataset contains the 1773 SFs, which had at least two nonredundant corresponding sequences in the PDB at the time of CATHe development. This SF dataset represented 87.6% of the CATH v4.3 database in terms of domain number. The smaller SF dataset is a subset of the bigger one, only keeping the most populated 50 SFs, representing 37.32% of the CATH v4.3 domain sequences in total. For each SF in the corresponding SF dataset, at least one domain sequence was placed in the test and validation set. Then, training sets were built using CATH-Gene3D sequences corresponding to the respective test and validation set SFs via scanning against CATH-HMMs [6]. In addition, sequences from PDB corresponding to the 3456 SF without two nonredundant sequences in PDB were included via the "other" part of the training dataset. All these other SF were under the single SF label "other" in the test and validation set, as there were not enough sequences in PDB to populate the test and validation set with each of these SF.

The sample sizes of the corresponding training datasets are 1 039 135 domains and 528 863 domains for the large and small datasets, respectively. As for the validation and testing sets, they both comprise 6863 and 1948 domains for the large and small datasets, respectively. For reference, CATH database contains 6573 SF at the time of writing, but most of them are admittedly very rare as in 2023, the 200 most highly populated superfamilies in CATH alone represent 62% of AF2 domains [7]. The CATHe dataset building process include one of the most stringent sequence identity filter in this context in order to assess the CATHe model. They made sure there was less than 20% sequence identity between and within the three sets.

Although the CATHe datasets were preserved as much as possible for CATHe2, some protein domains had to be removed during training and testing due to the incorporation of 3Di sequences. For the small dataset, there was no data loss. However, in the large dataset, the inability to retrieve certain PDB files required for computing 3Di led to the exclusion of 137 698 domains and 32 superfamilies from the original 1 039 135 and 1773, respectively. These removed superfamilies were among the least represented in the training set, comprising only 188 domains in total (Supplementary Fig. S1, see online supplementary material for a color version of this figure; Supplementary Table S2.) As discussed in the "Discussion" section, the removal of these 32 superfamilies has a negligible effect on overall performance. However, we emphasize that aside from losing these 32 classes, the training set also lost 137 510 domains (137 698—188), which spanned many of the remaining superfamilies. This reduction amounts to over 13% of the dataset used to train CATHe2 compared to CATHe. In a dataset where no pair of sequences shares more than 20% sequence identity—meaning every sample is highly informative—this significant reduction may outweigh the marginal benefit of reducing the number of superfamilies by 32. Moreover, the "other" SF representing all other SF than the 1772 present in the CATHe main dataset was not included in the CATHe2 dataset, due to a lack of information on these domains. It is worth mentioning that in the experiment where only AA sequence was considered, the dataset of CATHe2 does not suffer any superfamily loss compared to CATHe except for the "other" SF.

Additional experiments

For the sake of further and advanced experiments, we have created some smaller instances of the CATHe datasets by applying two different filters resulting in domain and CATH SF loss, although none of these filters affected the number of SF lost in the final CATHe2 models. The first filter tested was the predicted local distance difference test (pLDDT) threshold filter. One of the most interesting features generated by AF2 is a confidence score, i.e., pLDDT [3]. Since the accuracy of the CSM directly affects the accuracy of the 3Di sequences and consequently the accuracy of the models, we have tried to remove as much as possible the "low quality" structures in the hope to improve the models. Since the 3Di sequences from CATHe training dataset are computed on CSM from AFDB, which provides a confidence score, namely the pLDDT score, we have computed the overall pLDDT of all domains to be considered in the filter. The pLDDT threshold filter removes domains from the training set for which the pLDDT is under a certain threshold (pLDDT thresholds tested for CATHe2 models are: 0, 4, 14, 24, 34, 44, 54, 64, 74, 84); the SF completely removed from the training set by this filter were also removed from the test and validation set. Needless to say, a high threshold for such a filter not only leads to loss of SFs but also the reduction of the size of each class representing SFs (Supplementary Table S1; Supplementary Fig. S2, see online supplementary material for a color version of this figure). The second filter tested was a support threshold. The size of classes (size called support of the classes, a class being an SF here) is not only quite unbalanced, but also some SF comprises extremely low number of domains (<10) (Supplementary Fig. S1, see online supplementary material for a color version of this figure). This affects the ability of the models to train on such SFs, therefore, a support threshold filter was tested to try removing low support SF and improve the classifier performance. Support thresholds tested for CATHe2 models were 0 and 10, where 0 refers to not applying the filter. The SF completely removed from the training set by this filter were also removed from the test and validation set. A rather high number of SF was lost with support threshold = 10 (these lost SF include 167 SF which had already a support inferior or equal to 10 in CATHe raw training set, and 5 SF for which support was lowered under 10 by unfound 3D structure for 3Di usage). For this reason, no models trained with support threshold 10 have made it to the final version of CATHe2 although performance metrics were higher, see section "Results" for more details. Figure 1 summarizes these two filtering processes (Supplementary Fig. S2 [see online supplementary material for a color version of this figure] shows training set size in domains and SF relatively to the different filter thresholds).

3Di processing

Structural alphabets (SA) are sets of clusters grouping similar 3D conformations of short strings of AA. As each cluster is associated with a letter, the set of clusters constitutes an alphabets. They are used to represent the full 3D shape of proteins in one dimension, by building a sequence of SA letters, allowing for various 1D processing of 3D information. Most SA, such as CLE [47], 3D-BLAST [48], and Protein Blocks (PB) [49], discretize the conformations of short stretches of usually 3–5 Cα atoms. The SA chosen for this study is called 3D interaction (3Di) [17]; it does not describe the backbone but rather tertiary interactions. The 20 states (or 20 letters) of the 3Di alphabet describe the geometric conformation of each residue, with its spatially closest residue. 3Di has three key advantages over traditional backbone

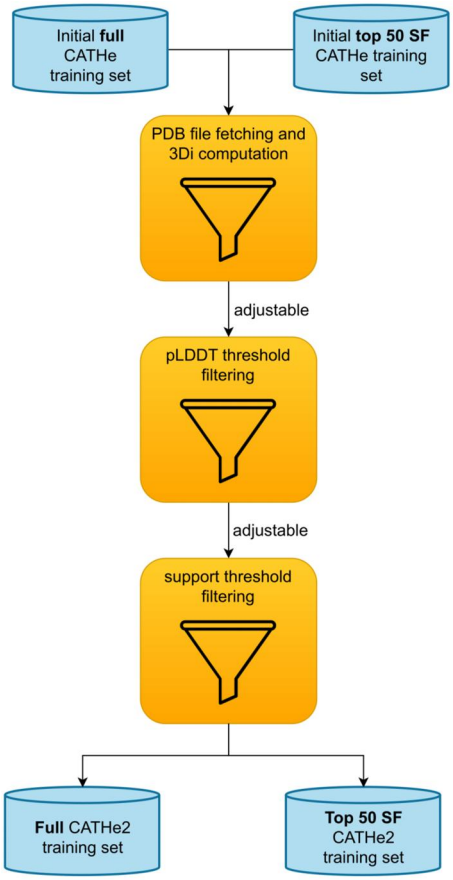


Figure 1 Overview of the dataset filtering process

Filtering process of CATHe2 datasets. Each filter is optional, pLDDT threshold and support threshold have adjustable values and many combinations were tested to improve model performance as much as possible without losing too much SF in the process.

structural alphabets. First, a weaker dependency between consecutive letters; second, more evenly distributed state frequencies, both enhancing information density and reducing false positives; third, the highest information density is encoded in conserved protein cores and the lowest in non-conserved coil/loop regions, whereas the opposite is true for backbone structural alphabets. In CATHe2, 3Di are computed using Foldseek [17] on PDB files from the RCSB Protein Data Bank (PDB) [50] for the test set and validation set. As the training set is composed of domains from the AlphaFold Protein Structure Database (AFDB) [4], Computed Structure Models (CSM) PDB files are extracted from there (when possible) to compute 3Di sequences for this dataset. See Fig. 2 for the whole CATHe2 input creation process.

Models

CATHe2 is an FNN classifier for which seven hyperparameters were fine tuned to achieve the best results possible. The first of these is the number of layer blocks, each block referring to a sequence of adjacent layers in the following order: a dense layer, a LeakyReLU (Leaky Rectified Linear Unit) activation layer, a batch normalization layer, and a dropout layer. The other hyperparameters were the dropout rate for the dropout layer, the dense layer size, the pLDDT threshold, the support threshold, the input type (AA embeddings only, 3Di embeddings only or the concatenation of both), and finally, the pLM used to embed AA sequences, which are presented in Table 2. It is to note that apart from ProtT5, the pLMs described below are all recent and were launched after the publication of CATHe paper.

Results

CATHe2 models performance

In order to have a comprehensive evaluation of different pLMs with different hyperparameters, we have run many preliminary experiments on CATHe's large dataset in order to come up with the ideal combination. To do this, some comparisons have been made with fixed classifier structure and dataset filters to compare all pLMs, and ProstT5 full was selected to fine tune other hyperparameters on (fine tuning all hyperparameters for each pLM would have been too computationally costly). Then a grid search has been performed to approach the best classifier structure and dataset filter values, guided by training loss analysis. Some more insights for this fine tuning and for the preliminary experiments are available in the supplementary material to this paper.

Based on the above, we have chosen the best combination found of all hyperparameters and filters to present a summary of the results. Although balanced accuracy and MCC were also computed for CATHe2 models, we have chosen to focus on F1 score since the dataset is highly unbalanced (95% confidence windows were computed using bootstrapping on the CATHe test set with 1000-folds and the standard deviation from this bootstrapping method was then multiplied with 1.96 to obtain the 95% confidence interval values). All F1 scores for all models tested for CATHe2 are available in the/results/perf dataframe.csv file in the GitHub project. Table 3 presents those results on the CATHe large dataset.

These results suggest ProstT5 full >ProstT5 half >ESM2 15B >ProtT5 >Ankh large >Ankh base >TM-Vec in the CATHe2 context, but it is important to note that each model and version of models can perform differently relative to the others, based on the other hyperparameters. Moreover, comparing the performance of different pLM embedding inputs in the same ANN classifier like in Table 3 is questionable as each pLM produces embeddings which are of different sizes and probably need different ANN classifier architecture to show maximum performance. However, ProstT5 full AA + 3Di embedding-based version of CATHe outperforms significantly other CATHe2 models and previous models, i.e., CATHe (F1 Score of 82.3% Vs 72.4%) [13]. As shown in Table 3, the fourth row, even with ProtT5 using our improved ANN algorithm and adjusted hyperparameters, the F1 Score didn't exceed 80.6%, which shows the added value of ProstT5 over ProtT5. The ProstT5 full AA only model available for inference in CATHe2 has the following hyperparameters: Nb Layer Block: 2, Dropout: 0.1 Input Type: AA only, Layer size: 1024, pLDDT threshold: 0, Support threshold: 0, and has an F1 score of 78.7%±1.3 on CATHe large dataset. The introduction of 3Di embeddings in addition to AA embeddings improved classifier performance (despite the loss of ∼13% of the training dataset compared to AA only), whereas 3Di embeddings on their own clearly underperform as shown in Fig. 3. This implies that AA sequences have more useful information for CATHe2 classifiers than 3Di sequences. This was expected since the classification of CATH's superfamilies is mainly based on sequence similarity, that is, evolutionary information. Not to add that the 3Di SA, similar to other structural alphabets, does not fully capture the structural information. To conclude this section, it is to note that

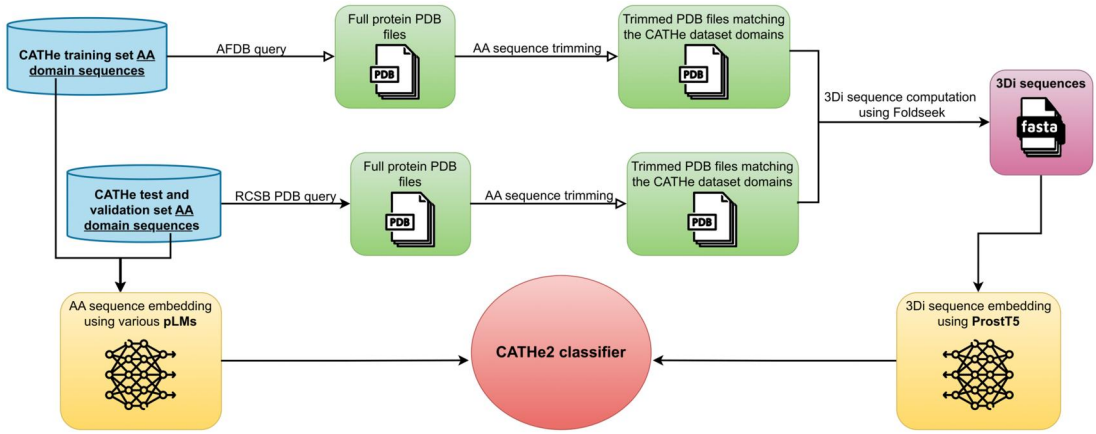


Figure 2 AA and 3Di sequence processing flowchart

CATHe datasets are composed of protein domains, not full protein sequences, therefore these domains had to be aligned to PDB and AFDB pdb file sequences, then trimmed, in order to keep only the 3D structure corresponding to the CATHe dataset domains. For this alignment, the Pairwise Aligner method from the Bio library was used [51]. These trimmed 3D structures were then used to compute 3Di sequences for ProstT5 to transform into 3Di sequence embeddings. Finally, combinations of 3Di embeddings and AA sequence embeddings were tested to train the CATHe2 classifier in order to get the highest F1 score possible. Meanwhile other hyperparameters than CATHe2 input type were tested like dataset filter values or classifier architecture.

Table 2. pLMs used for CATHe2 tests.

Model name	Input type	Training goal type	Training dataset	Specificities
ProtT5 (ProtT5-XL-U50) [14]	AA sequences only	Purely AA sequence based	BFD100 [3], UniRef50 [52]	Already used in CATHe and many other papers. Known to be performant. The oldest one.
ProstT5 (full and half precision) [26]	AA sequences or 3Di sequences	AA + 3D information oriented	AFDB [4]	Fine-tuned version of ProtT5. Can produce both AA and 3Di embeddings.
TM-Vec [45]	AA sequences only	3D information oriented	CATH, SwissModel [53]	Based on ProtT5 with extra layers to predict TM-scores [45].
ESM2 (version 15B, half precision) [43]	AA sequences only	3D information oriented	UniRef50, UniRef90 [52]	Very large model published in *Nature* alongside ESMFold [43].
Ankh large and Ankh base [22]	AA sequences only	AA sequence oriented	UniRef50 [52]	Inspired by the ProtT5 architecture. Focuses on protein-specific, empirical, knowledge-guided optimization. Ankh large is a larger version of Ankh base.

Presentation of the pLM used or CATHe2 AA and embeddings (and 3Di embeddings for ProstT5). Some more details on these pLMs are presented in the supplementary material file accompanying this paper, as well as the reasons why those were tested specifically.

Table 3. CATHe2 final models performance.

Embedding input for CATHe2	F1 score
ProstT5 full	82.3% ± 1.3%
ProstT5 half	81.9% ± 1.3%
ESM2 15B	80.6% ± 1.3%
ProtT5	80.6% ± 1.3%
Ankh large	78.3% ± 1.4%
Ankh base	77.3% ± 1.3%
TM-Vec	76.5% ± 1.4%

Hyperparameters corresponding to these results are the ones used for the best CATHe2 model, i.e., the ProstT5 full AA + 3Di embedding-based model on the large dataset without support threshold. These hyperparameters are Nb Layer Block: 2, Dropout: 0.3 Input Type: AA + 3Di, Layer size: 2048, pLDDT threshold: 24.0.

compared to the previous ANN classifier architecture, CATHe2 best models have more layers (2 layer blocks instead of 1) and larger dense layers (2048 nodes instead of 128). This increase of classifier complexity has been shown to improve results up to a certain point.

Additional experiments outcome

In this section, we will present the results of the advanced and further experiments, i.e., the pLDDT and support filters. As mentioned earlier, the combination of 3Di and AA inputs yielded consistently slightly better results than AA only trained models for the same hyperparameter combination, especially when applying a pLDDT threshold around 24 (see Fig. 4), whereas 3Di only models performed much worse consistently even with a high pLDDT threshold (see Fig. 3). The modest improvement on the F1 score of 3Di + AA over AA alone for most model tested (less than 1%)—aside from the fact that superfamily annotations rely heavily on evolutionary information typically embedded in amino acid sequences—may be attributed to the substantial ∼13% reduction in the training dataset when incorporating 3Di. Still, the best model of CATHe2 using both 3Di and AA outperforms the best model found using only AA by 3.3%. As for the pLDDT threshold influence, as Fig. 3 shows, high pLDDT thresholds resulted in poor F1 scores. The reason behind this is the loss of many domains in the training dataset. When applying a “strict” pLDDT filter (e.g. 84), the size of

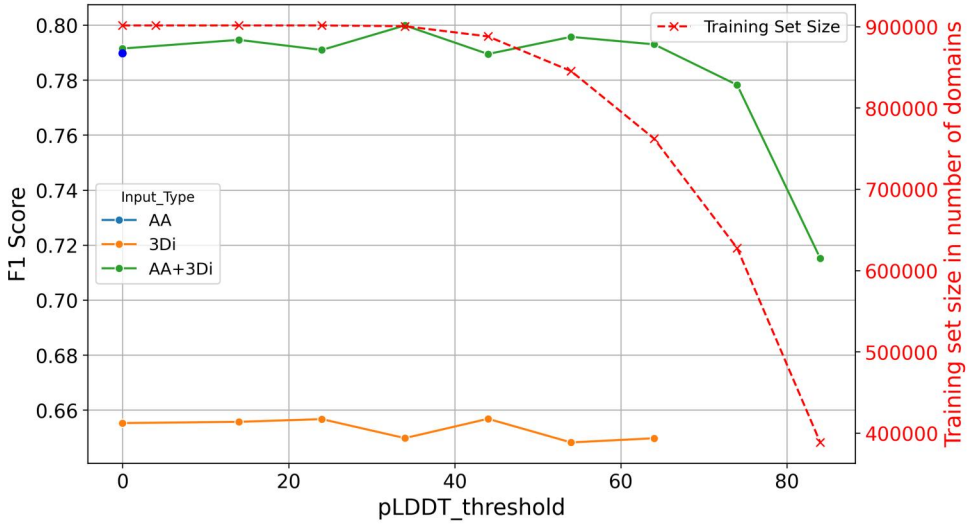


Figure 3 CATHe2 F1 score input type comparison along pLDDT threshold for a ProstT5 model

For AA only input there is only a single data point, for pLDDT threshold 0, because pLDDT is a 3D CSM confidence score, whereas no 3D models are used with only AA inputs. Moreover, the training set size mentioned for this AA only data point is not the one mentioned by the second y axis, but rather the full initial training dataset containing 1 039 135 domains. Hyperparameters for this model comparison are model embedding: ProstT5 full, Support threshold: 0, Nb layer block = 2, Layer size = 1024, Dropout = 0.3. Results are taken from CATHe2 large dataset, with no support threshold.

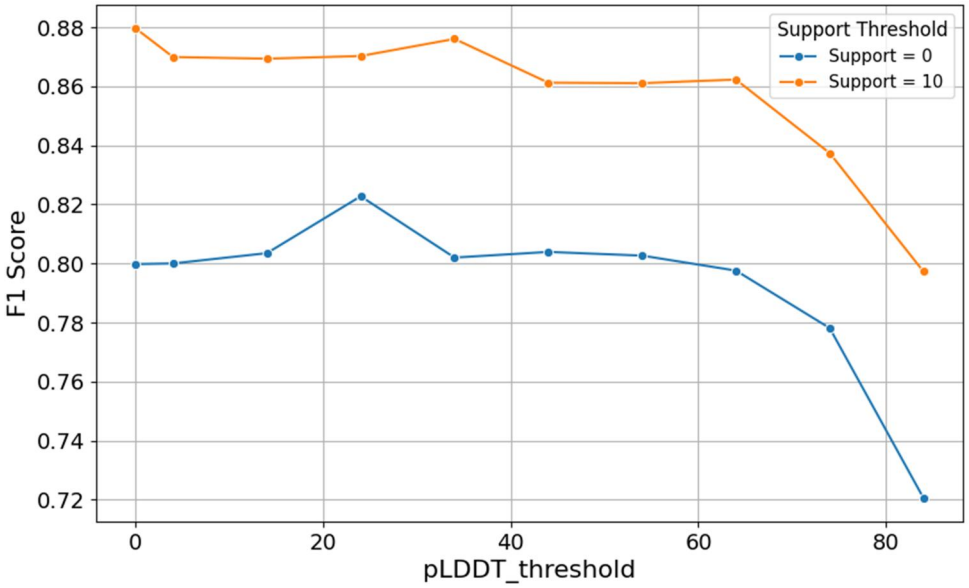


Figure 4 F1 score comparison for support threshold for the best hyperparameter combination

Contrary to the first plot, the hyperparameters here correspond to the best combination found for CATHe2 on its large dataset, i.e. ProstT5 full AA + 3Di embeddings with Dropout = 0.3, Layer size = 2048, and Nb layer block = 2.

each class/SF decreases significantly, which naturally affects the accuracy of the model. The conclusion here is that the negative effect of decreasing the size of the training dataset outweighs the positive effect of removing domains whose 3D structure is of medium quality. As for the second filter, i.e., the support filter, results are shown in Fig. 4. Due to a higher number of SF lost with support threshold (172 SF were lost with a threshold equal to 10), no models trained with it made it to the final version of CATHe2 despite achieving a very high performance of 88.0% ± 1% F1 score and accuracy of 93.8% ± 0.6% on the large dataset. This result shows again that training data sample size is very important to improve model performance and implies that a good part of future development should be focused on the gathering of more entries/domains for small SFs.

Discussion

Critical analysis of the study

In this paper, we proposed CATHe2, an updated version of CATHe (the CATH superfamilies automated classifier produced by the CATH team). Using roughly the same large dataset, we were able to attain an F1 score of 82.3% vs 72.4% (accuracy of 92.2% vs 85.6%). After a thorough study and comparison using different pLMs, hyperparameters, and methodologies, we propose the CATHe2 classifier with 2 layer blocks, 0.3 dropout rate for the dropout layers, and 2048 nodes for the dense layers. The input of this final model is a concatenation of ProstT5 AA sequence embeddings and ProstT5 3Di embeddings (pLDDT threshold of 24 for the CSM used to compute 3Di sequences). Although the CATHe2

dataset is quite large and is representative of the most populated SFs (the 200 most highly populated superfamilies in CATH alone represent 62% of AF2 domains [7]), CATHe2—similar to CATHe is bounded to infer only CATH superfamilies present in these ~1700 SF and therefore cannot predict a superfamily that was not in the training set. However, CATHe2 lacks the SF labelled "other" that was present in CATHe, which results in incomplete coverage of existing SFs at inference time, a clear limitation of the CATHe2 system. Nevertheless, given the large number of known SFs (over 1700), it is possible in practice for the predicted confidence scores to fall below 0.9 across all classes, particularly for domains that do not resemble any superfamily on which the model was trained. Following the approach adopted in CATH-Alphaflow [8], these confidence scores can serve as an empirical indicator of the overall trust in CATHe2's predictions. In this context, a probability threshold (e.g. 0.9) provides a practical heuristic for filtering uncertain results, where low-confidence predictions are typically rejected as less reliable. This confidence-based filtering does not constitute a formal identification of "other" or unseen superfamilies but offers a pragmatic means of assessing prediction reliability within the closed-set framework of the model. A more principled solution could involve contrastive learning [21], whereby the model evaluates the similarity of a new domain to known superfamilies and assigns it to a default "other" category when no close match exists. Alternatively, future work could consider reintroducing the former "other" class from CATHe into the CATHe2 framework to enhance coverage of novel or out-of-distribution domains.

In CATHe2, the absence of the "other" class is likely to have contributed to better performance, but it is unlikely to be significant as a ProtT5 model trained with the same hyperparameter as CATHe, without the "other," SF has only an F1 score of 73.8% ± 1.3%, compared to the 72.4% ± 0.7% of CATHe and the 82.3% ± 1.3% of the final CATHe2 model. Beyond the removal of the "other" superfamily detection, we had to remove 32 SF (out of 1773), due to the absence of their corresponding structures. However, missing those SF from the model is likely to explain only a small part of CATHe2 F1 score improvement too as a classifier similar to the CATHe one, trained on AA sequences only, but without those 32 SF (and without the "other" class), improved by 1.5% only compared to the same model but trained with those SFs (still without the "other" class). Having demonstrated that the two minor flaws in CATHe2 affected performance only marginally, it is worth emphasizing that, in the final and best-performing CATHe2 model presented in this study (which incorporates both AA and 3Di), more than 13% of the training dataset had to be excluded. One could argue that this data loss largely compensates for the positive impact on performance of the two aforementioned flaws.

Avenues for improvement

In this part, some new techniques and avenues for improvements are proposed to help future research on CATH annotation automation. A first promising improvement lead would be to implement Light Attention [46] when computing pLM embeddings per protein, instead of the average pooling on residue embeddings like in CATHe and CATHe2. As always, training the classifier with a larger, well curated dataset is also expected to improve results; to make this easier, 3Di sequences could be inferred from AA sequence only using ProstT5 associated CNN [26] or ESM2 embeddings [40]. Training the pLMs again during classifier training, instead of using frozen weights, is another promising lead. It may orient embedding computation toward the specialized task of CATH annotation prediction and thus improving results [54], although back propagation training for specific task fine tuning is not always successful and can lead to catastrophic forgetting or overfitting [14]. For CATHe2, the concatenation of AA embeddings from one pLM and 3Di embeddings from ProstT5 was tested. To go further, multiple AA embeddings from multiple pLMs and the 3Di embeddings from ProstT5 could be concatenated (or even 3Di embeddings from multiple 3Di trained pLMs). Specially, concatenating AA embeddings from ProstT5 and ProtT5 as the ProstT5 paper suggests could improve performance [26]. Although 3Di sequences were used in this study, it may be interesting to test protein blocks (PB) [49] instead, as the 3Di alphabet was not developed to maximize remote homology search sensitivity, but to model tertiary interactions in protein structures [17]. Even when using 3Di, a bilingual model like ProstT5 may not be optimal. Thus, a potential improvement on CATHe2 would be to use AA sequence embeddings from an AA sequence specialized pLM like ProtT5 and a 3Di (or PB) sequence specialized pLM which, to our knowledge, has not been developed yet. The pLMs tested for CATHe2 were just a promising subset of all current pLMs suited for the task. Other promising pLMs exist; for instance, ProSST is a good candidate, as it is a hybrid AA-structure pLM that produces its own 3Di-like structural representation with an advanced structure quantization method and a better attention formulation to leverage the structure cues [55]. Finally, SaProt embeddings could be used to improve CATHe2 results too. Very similar to ProstT5, SaProt is based on ESM2 650M instead of ProtT5. For SaProt, AA and 3Di were fused to create a novel structural alphabet on which it was trained [56]. SaProt could be one of the best pLMs to introduce structural information as suggested by other researchers [37]. Many other pLMs could be tested, and extensive lists can be found in the literature [10, 11]. Finally, other avenues for improvement include using embeddings from another hidden layer than the last one, using embedding denoising [9] and using other classifier architectures like CNN classifiers or even a combination of CNN classifiers, which seem to be more efficient than the simple FNN classifier used for CATHe and CATHe2 [57].

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Author contributions

Orfeu Mouret (Conceptualization [supporting], Data curation [lead], Investigation [lead], Methodology [lead], Software [lead], Visualization [lead], Writing—original draft [lead]) and Jad Abbass (Conceptualization [lead], Supervision [lead], Writing—review & editing [lead])

Supplementary data

Supplementary data are available at *Biology Methods and Protocols* online.

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Data availability

The code is available on https://GitHub.com/Mouret-Orfeu/CATHe2. Datasets: https://doi.org/10.5281/zenodo.14534966

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT-4o in order to assist with phrasing and to correct spelling. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

1. Anfinsen CB. The formation and stabilization of protein structure. *Biochem J* 1972;**128**:737–49. https://doi.org/10.1042/bj1280737.
2. Anfinsen CB. Principles that govern the folding of protein chains. *Science* 1973;**181**:223–30. https://doi.org/10.1126/science.181.4096.223.
3. Jumper J, Evans R, Pritzel A *et al.* High accuracy protein structure prediction using deep learning. *Nature* 2021;**596**:583–9. https://doi.org/10.1038/s41586-021-03819-2.
4. Varadi M, Anyango S, Deshpande M *et al.* AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. Nucleic Acids Res 2022;**50**:D439–44. https://doi.org/10.1093/nar/gkab1061
5. Das S, Lee D, Sillitoe I *et al.* Functional classification of CATH superfamilies: a domain-based approach for protein function annotation. *Bioinformatics* 2015;**31**:3460–7. https://doi.org/10.1093/bioinformatics/btv398.
6. Sillitoe I, Bordin N, Dawson N *et al.* CATH: increased structural coverage of functional space. *Nucleic Acids Res* 2021;**49**:D266–73. https://doi.org/10.1093/nar/gkaa1079.
7. Bordin N, Sillitoe I, Nallapareddy V *et al.* Alphafold2 reveals commonalities and novelties in protein structure space for 21 model organisms. *Commun Biol* 2023;**6**:160. https://doi.org/10.1038/s42003-023-04488-9.
8. Waman VP, Bordin N, Alcraft R *et al.* CATH 2024: CATH-AlphaFlow doubles the number of structures in CATH and reveals nearly 200 new folds. *J Mol Biol* 2024;**436**:168551. https://doi.org/10.1016/j.jmb.2024.168551.
9. Kilinc M, Jia K, Jernigan RL. Major advances in protein function assignment by remote homolog detection with protein language models—a review. *Curr Opin Struct Biol* 2025;**90**:102984. https://doi.org/10.1016/j.sbi.2025.102984.
10. Fan W, Zhou Y, Wang S, *et al.* Computational protein science in the era of large language models (LLMs). arXiv: 2501.10282, https://arxiv.org/abs/2501.10282, 2025, preprint: not peer reviewed.
11. Wang M, Ren Z. Deep learning methods for protein representation and function prediction: A comprehensive overview, unpublished preprint. http://dx.doi.org/10.2139/ssrn.5096181, 2025.
12. Ruan W, Lyu Y, Zhang J, *et al.* Large language models for bioinformatics. arXiv : 2501.06271, https://arxiv.org/abs/2501.06271, 2025, preprint: not peer reviewed.
13. Nallapareddy V, Bordin N, Sillitoe I *et al.* CATHe: detection of remote homologues for CATH superfamilies using embeddings from protein language models. *Bioinformatics* 2023;**39**:btad029. https://doi.org/10.1093/bioinformatics/btad029.
14. Elnaggar A, Heinzinger M, Dallago C *et al.* ProtTrans: toward understanding the language of life through self-supervised learning. *IEEE Trans Pattern Anal Mach Intell* 2022;**44**:7112–27. https://doi.org/10. 1109/TPAMI.2021.3095381.
15. Zhang C, Wang M, Xu Z, *et al.* A systematic study of joint representation learning on protein sequences and structures. arXiv : 2303.06275. https://arxiv.org/abs/2303.06275, 2023, preprint: not peer reviewed.
16. Borujeni PM, Salavati R. Functional domain annotation by structural similarity. *NAR Genom Bioinform* 2024;**6**:lqae005. https://doi.org/10.1093/nargab/lqae005.
17. van Kempen M, Kim SS, Tumescheit C *et al.* Fast and accurate protein structure search with foldseek. *Nat Biotechnol* 2024;**42**:243–6. https://doi.org/10.1038/s41587-023-01773-0.
18. Steinegger M, Söding J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat Biotechnol* 2017;**35**:1026–8. https://doi.org/10.1038/nbt.3988.
19. Steinegger M, Meier M, Mirdita M *et al.* HH-suite3 for fast remote homology detection and deep protein annotation. *BMC Bioinformatics* 2019;**20**:473. https://doi. org/10.1186/s12859-019-3019-7.
20. Remmert M, Biegert A, Hauser A, Söding J. HHblits: lightning fast iterative protein sequence searching by HMM-HMM alignment. *Nat Methods* 2011;**9**:173–5. https://doi.org/10.1038/nmeth.1818.
21. Holm L. Dali server: structural unification of protein families. *Nucleic Acids Res* 2022;**50**:W210–W215. https://doi. org/10.1093/nar/gkac387.
22. Elnaggar A, Essam H, Salah-Eldin W, *et al.* Ankh: Optimized protein language model unlocks general-purpose modelling. arXiv : 2301.06568. https://arxiv.org/abs/2301.06568, 2023, preprint: not peer reviewed.
23. Kilinc M, Jia K, Jernigan RL. Improved global protein homolog detection with major gains in function identification. *Proc Natl Acad Sci USA* 2023;**120**: https://doi.org/10.1073/pnas.2211823120.
24. Heinzinger M, Littmann M, Sillitoe I *et al.* Contrastive learning on protein embeddings enlightens midnight zone. *NAR Genom Bioinform* 2022;**4**:lqac043. https://doi.org/10.1093/nargab/lqac043.
25. Bepler T, Berger B. Learning the protein language: evolution, structure, and function. *Cell Syst* 2021;**12**:654–69.e3. https://doi.org/10.1016/j.cels.2021.05.017.
26. Heinzinger M, Weissenow K, Sanchez JG *et al.* Bilingual language model for protein sequence and structure. *NAR Genom Bioinform* 2024;**6**:lqae150. https://doi.org/10.1093/nargab/lqae150.
27. Pantolini L, Studer G, Pereira J *et al.* Embedding-based alignment: combining protein language models with dynamic programming alignment to detect structural similarities in the twilight-zone. *Bioinformatics* 2024;**40**:btad786. https://doi.org/10.1093/bioinformatics/btad786.
28. Iovino BG, Ye Y. Protein embedding based alignment. *BMC Bioinformatics* 2024;**25**:85. https://doi.org/10.1186/s12859-024-05699-5.
29. Kaminski K, Ludwiczak J, Pawlicki K *et al.* pLM-BLAST: distant homology detection based on direct comparison of sequence representations from protein language models. *Bioinformatics* 2023;**39**:btad579. https://doi.org/10.1093/bioinformatics/btad579.
30. Erckert K, Rost B. Assessing the role of evolutionary information for enhancing protein language model embeddings. *Sci Rep* 2024;**14**:20692. https://doi.org/10.1038/s41598-024-71783-8.
31. Lupo U, Sgarbossa D, Bitbol A-F. Protein language models trained on multiple sequence alignments learn phylogenetic

relationships. *Nat Commun* 2022;**13**:6298. https://doi.org/10.1038/s41467-022-34032-y.
32. Gong C, Klivans A, Loy JM, *et al.* Evolution-inspired loss functions for protein representation learning. In: *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, Vol. 235. JMLR.org, Article 637, pp. 15893–15906. Vienna, Austria, 2024. https://dl.acm.org/doi/10.5555/3692070.3692707
33. Rao J, Liu R, Verkuil RM, *et al.* MSA transformer. In: *Proceedings of the 38th International Conference on Machine Learning, Vol. 139 of Proceedings of Machine Learning Research, PMLR*, 2021, pp. 8844–8856. Virtual event. https://proceedings.mlr.press/v139/rao21a.html
34. Labiosa A, Kohli C, Chung C, Korban N. Hybrid transformer and neural network configuration for protein classification using amino acids. bioRxiv. https://doi.org/10.1101/2024.10.31.621421, 2024.
35. Gligorijevic' V, Renfrew PD, Kosciolek T *et al.* Structure-based protein function prediction using graph convolutional networks. *Nature Communications* 2021;**12**:6101. https://doi.org/10.1038/s41467-021-23303-9.
36. Ghazikhani H, Butler G. Toot-plm-p2s: incorporating secondary structure information into protein language models. *Unpublished Preprint* 2024; https://doi.org/10.1101/2024.08.13.607781.
37. Frolova, D Pak, MA Litvin, A *et al.* Mulan: Multimodal protein language model for sequence and structure encoding. https://doi.org/10. 1101/2024.05.30.596565, 2024, unpublished preprint.
38. Zhao J, Zhuang W, Song J *et al.* 2024. Pre-training protein bi-level representation through span mask strategy on 3D protein chains. In: *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, Vol. 235. JMLR.org, Article 875, 21791–21805. https://dl.acm.org/doi/10.5555/3692070.3692945
39. Schütze K, Heinzinger M, Steinegger M, Rost B. Nearest neighbor search on embeddings rapidly identifies distant protein relations. *Frontiers in Bioinformatics* 2022;**2**:1033775. https://doi.org/10.3389/fbinf.2022.1033775.
40. Johnson SR, Peshwa M, Sun Z. Sensitive remote homology search by local alignment of small positional embeddings from protein language models. *eLife* 2024;**12**:RP91415. https://doi.org/10.7554/eLife.91415.
41. Zhang Y, Zhang W. Rprot-Vec: A deep learning approach for fast protein structure similarity calculation. *BMC Bioinformatics* 2025;**26**:171. https://doi.org/10.1186/s12859-025-06213-1.
42. Lin B, Luo X, Liu Y, Jin X. A comprehensive review and comparison of existing computational methods for protein function prediction. *Briefings in Bioinformatics* 2024;**25**:bbae289. https://doi.org/10. 1093/bib/bbae289.
43. Lin Z, Akin H, Rao R *et al.* Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 2023;**379**:1123–30. https://doi.org/10.1126/science.ade2574.
44. Evans R, O'Neill M, Pritzel A *et al.* Protein complex prediction with alphafold-multimer. *bioRxiv*, https://doi.org/10.1101/2021.10.04.463034, 2022, preprint: not peer reviewed.
45. Hamamsy T, Morton JT, Blackwell R *et al.* Protein remote homology detection and structural alignment using deep learning. *Nat Biotechnol* 2024;**42**:975–85. https://doi.org/10.1038/s41587-023-01917-2.
46. Stärk H, Dallago C, Heinzinger M, Rost B. Light attention predicts protein location from the language of life. *Bioinformatics Advances* 2021;**1**:vbab035. https://doi.org/10.1093/bioadv/vbab035.
47. Wang S, Zheng W-M. Clepaps: fast pair alignment of protein structures based on conformational letters. *J Bioinform Comput Biol* 2008;**06**:347–66. https://doi. org/10.1142/S0219720008003461.
48. Yang J-M, Tung C-H. Protein structure database search and evolutionary classification. *Nucleic Acids Res* 2006;**34**:3646–59. https://doi.org/10.1093/nar/gkl395.
49. de Brevern A, Etchebest C, Hazout S. Bayesian probabilistic approach for predicting backbone structures in terms of protein blocks. *Proteins: Structure, Function, and Bioinformatics* 2000;**41**:271–87. https://doi.org/10.1002/1097-0134(20001115)41:3<271::AID-PROT10>3.0.CO;2-Z.
50. Berman H, Westbrook J, Feng Z *et al.* The protein data bank. *Nucleic Acids Research* 2000;**28**:235–42. https://doi.org/10.1093/nar/28.1.235. URL https://www.rcsb.org/
51. Cock PJA, Antao T, Chang JT *et al.* Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 2009;**25**:1422–3. https://doi.org/10.1093/bioinformatics/btp163
52. Suzek BE, Wang Y, Huang H, the UniProt Consortium *et al.* UniRef clusters: a comprehensive and scalable alternative for improving sequence similarity searches. *Bioinformatics* 2015;**31**:926–32. https://doi.org/10.1093/bioinformatics/btu739.
53. Waterhouse A, Bertoni M, Bienert S *et al.* Swiss-model: homology modelling of protein structures and complexes. *Nucleic Acids Res* 2018;**46**:W296–W303. https://doi.org/10.1093/nar/gky427.
54. Schmirler R, Heinzinger M, Rost B. Fine-tuning protein language models boosts predictions across diverse tasks. *Nat Commun* 2024;**15**:7407. https://doi.org/10.1038/s41467-024-51844-2.
55. Li M, Tan P, Ma X, *et al.* ProSST: Protein language modeling with quantized structure and disentangled attention. Unpublished preprint. https://doi.org/10.1101/2024.04.15.589672, 2024.
56. Su J, Han C, Zhou Y, *et al.* SaProt: Protein language modeling with structure-aware vocabulary. Unpublished preprint. https://doi.org/10.1101/2023.10.01.560349, 2024.
57. Vitale R, Bugnon LA, Fenoy EL *et al.* Evaluating large language models for annotating proteins. *Brief Bioinform* 2024;**25**:bbae177. https://doi.org/10.1093/bib/bbae177.

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